A loop with teeth

The chain loops around a front chainring and one back cog. When the pedals turn forward, the moving chain helps turn the rear wheel.

Cable and spring

A shifter pulls or releases a cable. The rear derailleur swings sideways and guides the moving chain toward the next back cog.

Two pulley jobs

The upper pulley guides the chain. The lower pulley and a spring take up extra chain, keeping the long loop snug.

The front cage

On bikes with several front chainrings, a metal cage nudges the moving chain sideways from one chainring to another.

Keep it moving

The chain must move forward to climb onto another cog. A shifter click alone cannot slide a still chain into place.

Easy or hard

A big back cog makes climbing easier. A small back cog turns the wheel farther per pedal turn but needs more push.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How Bike Derailleurs Work